

CARNEGIE MELLON UNIVERSITY
DATA, INFERENCE & APPLIED MACHINE LEARNING (COURSE 18-785)
ASSIGNMENT 4

INSTRUCTIONS

- Submissions should be made via Canvas.
- **Single Python/MATLAB code file(.ipynb or .m) [Do not submit checkpoints for .ipynb].** In addition, each line of code should be documented by text. This demonstrates that the code is unique and owned by the student.
- Assignment report(.pdf) with full evidence that the assignment was completed by the student and demonstrates a full understanding of each step in the process, including textual descriptions of each result (statistics, table, graph etc.) and insights that can be gained.
- Indicate the libraries you have used in your code at the beginning of the report (After the title page).
- Using ChatGPT for any assignment is not allowed, as it could lead to being flagged for plagiarism.
- Data files (as given).

Submission process:

1. Put the source code **file and data files** in a single folder
2. The name of the folder should be the same as your Andrew ID
3. **Zip this folder and attach the zipped file on the assignment submission page (CANVAS)**
4. After attaching the zipped file, click on "Add Another File" from the assignment submission page and **attach your report**
5. Submit your assignment

N.B. This process will allow us to compile your reports in **Turnitin** to check for plagiarism.

Specific reasons for a submission being classified as incomplete include:

- Failure to correctly name your folder with your Andrew ID
- Failure to correctly name your report, and code file with andrewID_DIAML_AssignmentNo. For example, mcsharry_DIAML_Assignment1, mcsharry_DIAML_Assignment2, and mcsharry_DIAML_Assignment3.
- A missing report describing the steps, results, and insights
- A missing dataset required for running the code
- A missing code file such as a .ipynb or .m file
- An error in the file path needed to run the code

The student is responsible for checking that their submission is complete. Students will lose 10% as for late submission even if the submission is repaired during the 24 hours after the deadline has passed, and receive 0 for the assignment if it is not repaired.

The submission deadline is **on Monday, 12, October 2026, 17:59 Eastern Time (ET) /**

Monday 12, October, 2026 23:59 Rwandan Time (CAT).

Q1. Linear regression with one explanatory variable((20 points)

Load [monthly house price data](#) and the [FTSE100 index data](#), restricted to January 1991 through August 2026.

- (a) Fit a linear regression using the raw Average House Price level as the explanatory variable and the raw FTSE100 Close level as the dependent variable. Then compute the monthly return of each series as $r(t) = p(t)/p(t-1) - 1$, and fit the same regression using returns instead of levels. For both regressions, report the slope, intercept, correlation coefficient, R^2 , and p-value.
- (b) These two regressions give strikingly different answers. Explain why, by checking how much of each series's own variation is explained purely by the passage of time.
- (c) Interpret the slope, intercept, correlation, and R^2 from the returns-based regression. What does each one tell you about the relationship between house price returns and FTSE returns? Use a hypothesis test to back up your conclusion about whether a significant relationship exists.
- (d) Slope and correlation are not the same thing; explain what each one is actually telling you, and why a small regression slope can coexist with a strong correlation, and vice versa. Plot the return data points together with the fitted regression line, and discuss what the plot shows in light of your answer.
- (e) Choosing the dependent and explanatory variables in a regression does not establish causality; this model treats house-price returns as explaining FTSE returns. Does the data alone justify that direction, or could the roles of x and y just as easily be reversed? Rerun the regression with the two variables swapped and report the new slope, intercept, correlation, and R^2 . Which of these changes, and which stays the same? And why?

Q2. Linear Regression with Multiple Explanatory Variables((30 points))

The [College file](#) contains information about different US colleges and universities. We will use the number of applications received, the number of enrolled students, the out-of-state tuition, the percentage of new students who were in the top 10% of their high school class, the percentage of new students who were in the top 25% of their high school class, the percentage of faculty holding a PhD, the percentage of faculty holding a terminal degree, and the instructional expenditure per student to predict the graduation rate.

Use Apps, Enroll, Outstate, Top10perc, Top25perc, PhD, Terminal, and Expend to predict Grad.Rate.

- (a) Examine the distributions of both the dependent and independent variables, identify any unusual values, and explain how you handle them before fitting the models.
- (b) What is multicollinearity, and why is it a problem for a regression with multiple predictors? Compute the correlation matrix for all eight predictors and identify any strongly correlated pairs. Then compute the Variance Inflation Factor (VIF) for each predictor. Do the pairs you flagged from the correlation matrix match the predictors with $VIF > 5$? If not, explain the discrepancy and explain why VIF can detect multicollinearity that is not apparent from pairwise correlations.
- (c) After removing the worst multicollinear predictor, use stepwise regression to build linear regression models via forward(entry $p < 0.05$) and backward(removal $p > 0.10$) selection on the remaining seven predictors. Report which predictors both directions agree on, and which they disagree on, and explain why. Which model has the better adjusted R^2 and cross-validated RMSE using 10 folds?
- (d) Compare the BIC of forward's model, backward's model, and the best-BIC model across all subsets of the seven predictors. Do BIC and your Part C accuracy comparison agree on which model is best?
- (e) Fit a full model using all seven remaining predictors and compute its adjusted R^2 and cross-validated RMSE using 10 folds. Compare all four models: forward's, backward's, the best-BIC model from part D, and this full model, and identify the most accurate one.

Q3: Open study ((20 points))

Design and undertake a study to assess a trend in the domain of transport for one or more countries of your choice. Your study should be based on publicly available data and explained using mathematical facts. Explain assumptions, methodology, and findings. An example would be to study the relationship between an increase in transport and road traffic accidents. The World Health Organization has data for road traffic deaths per country, and there is a World Bank indicator for Passenger cars (per 1000 people). Can historical data help predict changes in road traffic deaths? Evaluate your prediction using observations not included in model fitting.

Deliverables: You should turn in a report that includes the trend you are studying, the data source, your assumptions, the methodology used along with its implementation in MATLAB/Python, and finally the findings and conclusions, which should be backed with code and figures.

Q4: Model Fitting and Prediction((30 points))

By using the Dataset of the [annual unemployment rate for Israel](#), World Bank indicator SL.UEM.TOTL.NE.ZS.

(a) Plot the full series. Describe, in your own words, the distinct phases visible in the data(periods of sustained rise and periods of sustained decline)

(b) Fit a linear regression of the unemployment rate on year using the entire dataset, then evaluate that same model by predicting every year in the dataset. Report the RMSE or MAE of these predictions.

Plot the actual vs. predicted. Explain why the accuracy figure is not a trustworthy way to judge how well this model would perform on data it hasn't already seen.

(c) Given the shape you described in (a), a straight line may not be the right form. Fit a quadratic regression, and evaluate it using the same metric as (b). Plot the actual vs. predicted. Does the model improve accuracy compared to the straight-line fit?

(d) Reserve the final ten available years as a test set. Fit the linear and quadratic models using only the preceding observations. Predict the ten test years and calculate the RMSE for each model."

(e) Now compute a naive baseline for the same test portion: predict every year using nothing but the last known value from the training portion. Report its RMSE or MAE and compare it to both the linear (b) and quadratic (c) models, as evaluated in (d). Plot the actual vs. predicted. Which of the three performs best?

(f). Using the same training endpoint as in (e), report the absolute forecast errors of the naive model at horizons of 1, 2, and 3 years. Does it stay the same, improve, or get worse as the horizon grows? What does this tell you about how far into the future a naive-persistence estimate can reasonably be trusted?

(g) State which of the three approaches(linear, quadratic, or naive) you would actually use to estimate the unemployment rate of Israel for 2026, and report that estimate using the refitted model on all available observations.